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NOTES:

A. MATERIAL AND FINISH:
TOP HOUSING – ALUMINUM
BOTTOM HOUSING – DIE CAST ZINC
CAM, TRIGGER – SINTERED METAL, STEEL, ZINC PLATED
SPRINGS – STAINLESS STEEL, PASSIVATED
TRIGGER INTERLOCK LEVER – GLASS FILLED NYLON
BISTABLE SPRING RETAINER – DIE CAST ZINC
DRIVE CAM – BLACK ACETAL
OUTPUT CAM – WHITE ACETAL

B. ELECTRICAL SPECIFICATIONS:
OPERATING VOLTAGE: 12-24 VDC NOMINAL
TYPICAL OPERATING CURRENT: LESS THAN 500 mA AT 12 VDC
PEAK / STALL OPERATING CURRENT: 1 A MAX (STALL LIMITED TO 1.4 SECONDS)
TOTAL STANDBY CURRENT, LOCKED / UNLOCKED: 5 mA
CONTROL SIGNAL HIGH (UNLOCK COMMAND): 6 VDC TO OPERATING VOLTAGE, 5.5 mA MAX
CONTROL SIGNAL LOW (LATCHED COMMAND): 0 TO 1 VDC (OPEN)
LATCH TRANSIT TIME TO RELEASE: 600 MILLISECONDS NO LOAD, 1.4 SECOND MAX
OPERATING TEMPERATURE RANGE: -20C TO 50C

C. ELECTRICAL CONNECTIONS AND HOOKUP:
A BASIC SWITCH CONTROL ELECTRICAL HOOKUP DIAGRAM IS PROVIDED FOR REFERENCE
CONSULT WITH A SOUTHCO REPRESENTATIVE FOR ADDITIONAL ELECTRICAL HOOKUP INFORMATION
- CONNECT POWER, GROUND AND CONTROL SIGNAL WIRES TO AN APPROPRIATE DC POWER SUPPLY
- A DC POWER SUPPLY CAPABLE OF SUPPLYING 12-14 VDC AT 1 AMP MINIMUM PER LATCH IS RECOMMENDED
- POWER MUST BE AVAILABLE TO OPERATE THE LATCH AND MUST REMAIN AVAILABLE DURING THE FULL TRANSIT TIME OF THE LATCH DURING LOCKING OR UNLOCKING.
CAUTION! LATCH CAN BE DAMAGED IF WIRED INCORRECTLY, OR IF IMPROPER VOLTAGE IS APPLIED!
CONNECTOR PIN ASSIGNMENT: SEE CONNECTOR PINOUT TABLE AND PIN LOCATION DETAILS.
STATUS INDICATION: OPEN COLLECTOR OUTPUT. RATING 12-24 VDC. 50 mA MAX LOAD.

D. ELECTRICAL OPERATION:
TO UNLOCK OR RELEASE THE LATCH:
PROVIDE THE FOLLOWING CONTROL SIGNAL TO CONNECTOR PIN 3
- PROVIDE 12-24 VDC (CONTROL SIGNAL HIGH) FOR A MINIMUM OF 50 MILLISECONDS
- THE CONTROL SIGNAL CAN REMAIN HIGH INDEFINITELY
- THE DELAYED RE-LOCK LATCH WILL STAY UNLOCKED FOR A MINUMUM OF 1 MILLISECOND OR AS LONG AS THE CONTROL SIGNAL IS HIGH.
- THE AUTO RE-LOCK LATCH WILL STAY UNLOCKED FOR A MINIMUM OF 1 MILLISECOND AND RE-LOCK IF THE CONTROL SIGNAL REMAINS HIGH.
TO LOCK THE DELAYED RE-LOCK LATCH AND TO ENABLE THE AUTO RE-LOCK LATCH FOR A NEW OPENING CYCLE:
PROVIDE THE FOLLOWING CONTROL SIGNAL TO CONNECTOR PIN 3
- PROVIDE CONTROL SIGNAL LOW FOR 50 MILLISECONDS. POWER MUST BE AVAILABLE DURING TRANSIT TO THE LOCKED POSITION.

E. OUTPUT FEEDBACK:
- OPEN COLLECTOR OUTPUT. RATING 12-14VDC. 50 mA MAX LOAD
CAUTION! TO PREVENT DAMAGE TO THE PRODUCT DO NOT EXCEED MAXIMUM LOADS STATED AND FOLLOW WIRING DIRECTIVES.

F. LATCH CONNECTOR:
MANUFACTURER: MOLEX, SERIES: MICRO-FIT 3.0
-3.0mm PITCH MICRO-FIT HEADER, DUAL ROW, 6 POSITIONS, MOLEX P/N: 43045-0606 OR EQUIVALENT.
MATE CONNECTOR / WIRE HARNESS REQUIRED (NOT SUPPLIED)
- CONNECTOR: RECEPTACLE HOUSING, DUAL ROW, 6 POSITION 3mm: MOLEX P/N: 43025-0600
- CONTACTS: FEMALE CRIMP TERMINAL (SOCKET) MOLEX P/N 43030-0007

G. MOUNTING:
- MOUNT THE LATCH SECURELY USING TWO (2) SCREWS IN THE MOUNTING HOLES PROVIDED (SCREWS NOT PROVIDED)
- MOUNTING CONFIGURATIONS ARE AVAILABLE WITH THREADED OR THROUGH HOLES PER ASSEMBLY PART NUMBER SCHEME OUTLINED ON SHEET 2
- SEE SHEET 3 AND SHEET 4 FOR SUGGESTED MOUNTING CONFIGURATIONS AND HARDWARE REQUIREMENTS

H. MECHANICAL OPERATION:
THE LATCH IS PROVIDED WITH A MECHANICAL TRIGGER TO RELEASE THE LATCH.
THE MINIMUM TRAVEL REQUIRED TO RELEASE THE LATCH IS SHOWN ON SHEET 2.
THE TRIGGER MOVES THROUGH ITS FULL TRAVEL DURING ELECTRICAL OPERATION OF THE LATCH.
ACCESSORIES ARE AVAILABLE (SOLD SEPARATELY) FOR REMOTE ACTUATOR CABLE INSTALLATION (SEE DRAWING J-R4-EM-952 FOR MANUAL OVERRIDE BRACKET).
CAUTION! TO PREVENT LONG TERM DAMAGE OF ELECTRICAL COMPONENTS IN THE LATCH, DO NOT OBSTRUCT MOTION OF THE TRIGGER DURING ELECTRICAL OPERATION

I. STRIKER ASSEMBLY SOLD SEPARATELY
REFER TO J-R4-90-709-20 FOR ADDITIONAL INFORMATION ON STAINLESS STEEL STRIKER
AND J-R4-90-804-10 FOR ADDITIONAL INFORMATION ON ZINC DIE CAST STRIKER ASSEMBLY.

J. KICK OUT FORCE UPON UNLOCKING FROM CLOSED POSITION: (5N)
HOLDING FORCE UPON UNLOCKING FOR PULL-TO-OPEN LATCH: (15N)

K. PACKAGED IN INDIVIDUAL BOXES. ADD -1 SUFFIX TO PART NUMBER FOR BULK PACKAGED OPTION.

REVISION HISTORY

REV	DATE	BY	DESCRIPTION
F	21MAY2024	CRM/DMS	PRN: P2024-0871

EXTERNAL TO LATCH

INTERNAL TO LATCH

OUTPUT OPEN DRAIN (PIN 4,6)

GROUND (PIN 1)

OUTPUT FEEDBACK

MICRO PROCESSOR

1 GROUND

2 +12 TO 24VDC

3 +CONTROL SIGNAL

4 LATCH STATUS (OPEN DRAIN)

5 NONE

6 DOOR STATUS (OPEN DRAIN)

SCALE 1:2

GROUND

+12 TO 24 VDC

POWER SUPPLY

CONTROL SWITCH

R4-EM LATCH

GROUND

+12 TO 24VDC

CONTROL SIGNAL (IN)

LATCH STATUS

NONE

DOOR STATUS

ELECTRICAL HOOKUP (OUTPUT/FEEDBACK)

CONNECTOR PINOUT	
PIN	FUNCTION
1	GROUND (-)
2	POWER (+)
3	CONTROL SIGNAL
4	LATCH STATUS
5	NONE
6	DOOR STATUS

CONNECTOR PINOUT		OUTPUT	
PIN	FUNCTION	OPEN	GROUND
4	LATCH STATUS	LOCKED	UNLOCKED
5	NONE	NOT APPLICABLE	
6	DOOR STATUS	CLOSED	OPEN

SEE SHEET 2 OF 2 FOR LATCH PART NUMBERS AND INSTALLATION INFORMATION

	THIRD ANGLE PROJECTION	southco® CONNECT • CREATE • INNOVATE			
	MILLIMETERS [IN]	ELECTRONIC ACCESS SOLUTION R4-EM 9 SERIES DIE CAST			
SURFACE AREA N/A mm ²	TOLERANCES UNLESS OTHERWISE NOTED ALL DIMENSIONS WITHOUT TOLERANCES ARE FOR REFERENCE ONLY.	SIZE A3	SYSTEM NX	DWG NO. J-R4-EM-9-2	
VOLUME N/A mm ³					
PROPRIETARY ITEM EXCEPT FOR USES EXPRESSLY GRANTED IN WRITING, INFORMATION DISCLOSED HEREON IS CONFIDENTIAL AND ALL RIGHTS, PATENT AND OTHERWISE, ARE RESERVED BY SOUTHCO, INC.					
	PER ASME Y14.5M-2009	DRAWN BY BVH/	DATE 06APR2017	SCALE 1:1	SHEET 1 OF 4

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REVISION HISTORY			
REV	DATE	BY	DESCRIPTION
F	21MAY2024	CRM/DMS	PRN: P2024-0871

ASSEMBLY PART NUMBER

R4-EM-TBA5-1SC-P

'T' DESIGNATES TRIGGER STYLE
9A = AUTO RE-LOCK KICK OUT
9D = DELAYED RE-LOCK KICK OUT
9P = DELAYED RE-LOCK PULL TO OPEN

'B' DESIGNATES PIVOT PIN TYPE
1 = 1/4-20
2 = M6
3 = 7MM THRU

'P' DESIGNATES PACKAGING
BLANK = INDIVIDUAL
1 = BULK
'C' = 0 - NO EXTERNAL CONNECTOR

'S' DESIGNATES SENSOR OPTIONS
5 = DOOR AND TRIGGER SENSOR (STANDARD OUTPUT)
EXTENDED HOUSING ONLY (A = BLANK)
5A = DOOR AND TRIGGER SENSOR (REVERSED OUTPUT)
EXTENDED HOUSING ONLY (A = BLANK)

'A' DESIGNATES ALTERNATIVE CONFIGURATIONS
BLANK = EXTENDED HOUSING HEIGHT

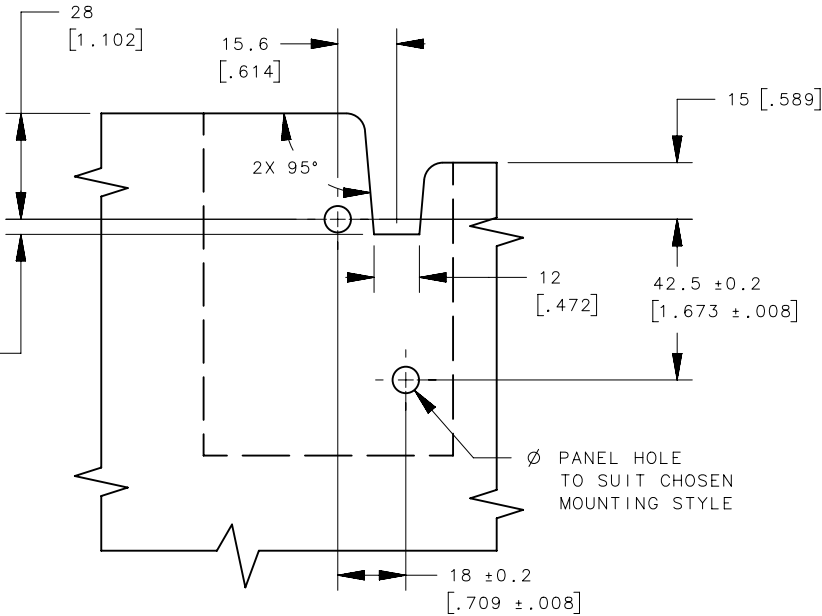
STRIKER
R4-90-709-20 (SHOWN)
R4-90-804-10
SOLD SEPARATELY

MANUAL OVERRIDE
BRACKET
R4-EM-952
SOLD SEPARATELY

STRIKER IN UNLATCHED STATE
SCALE 1:2

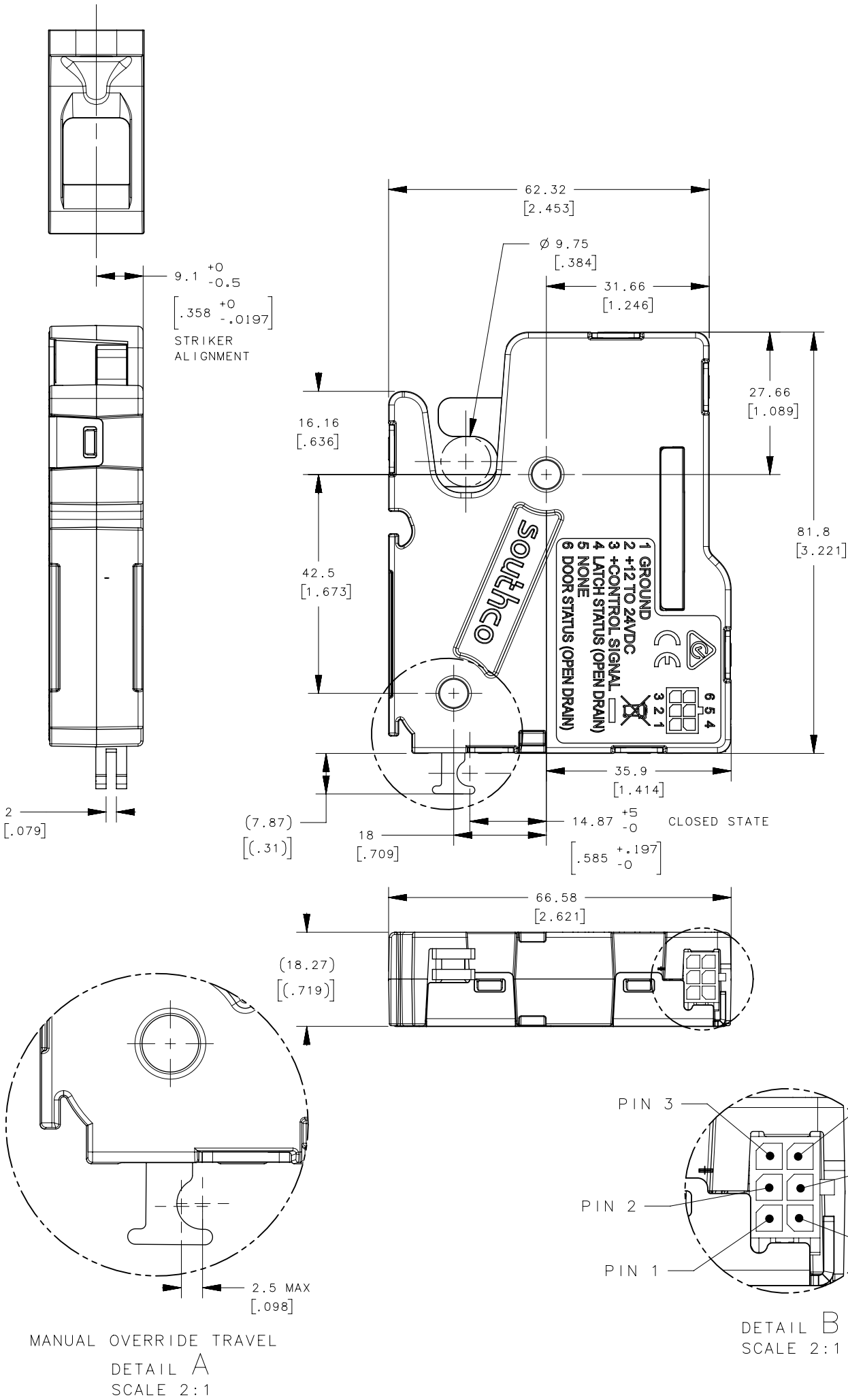
37.2
[1.465]
STRIKER IN
CLOSED POSITION

STRIKER IN LATCHED STATE
SCALE 1:2



PANEL PREPARATION
SCALE 1:2

37.2 +10
-1
[1.465 +.394
-.039]
DOOR STATUS
'DOOR CLOSED'
POSITION



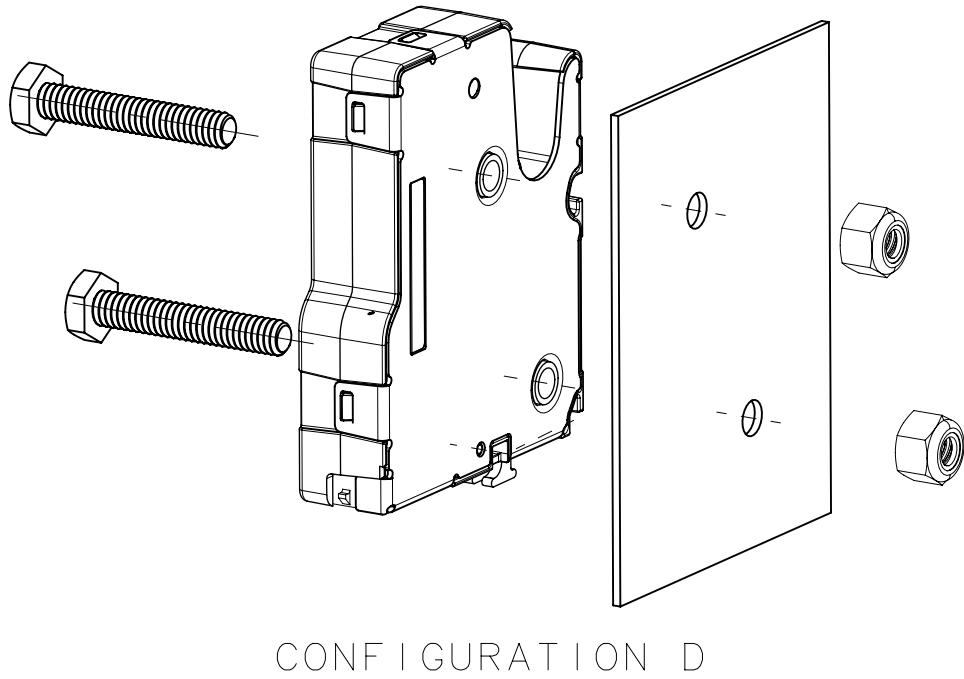
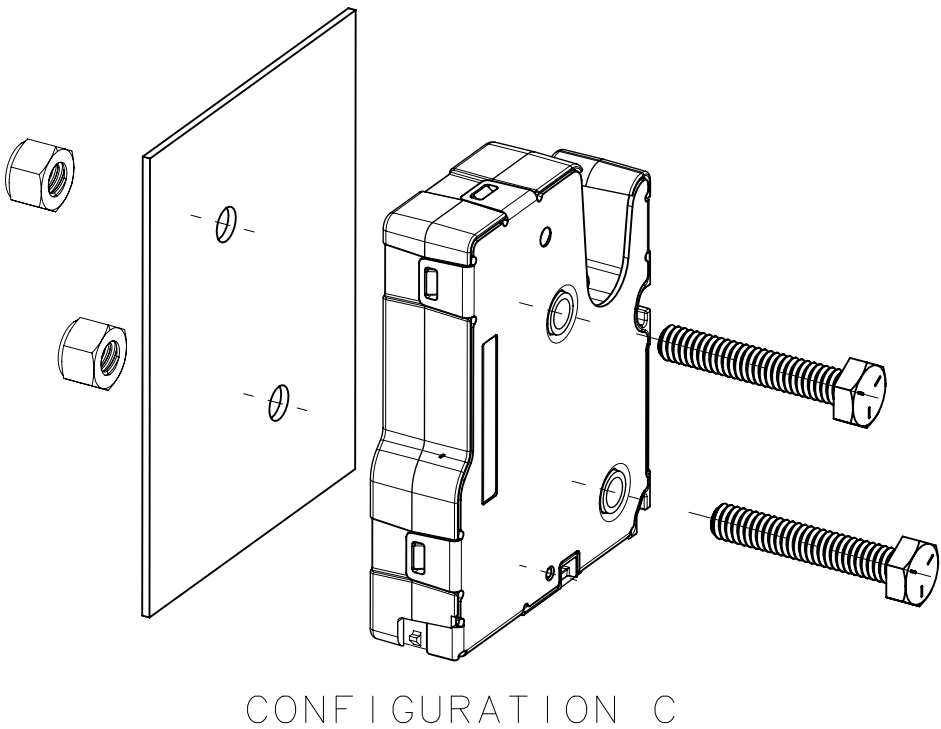
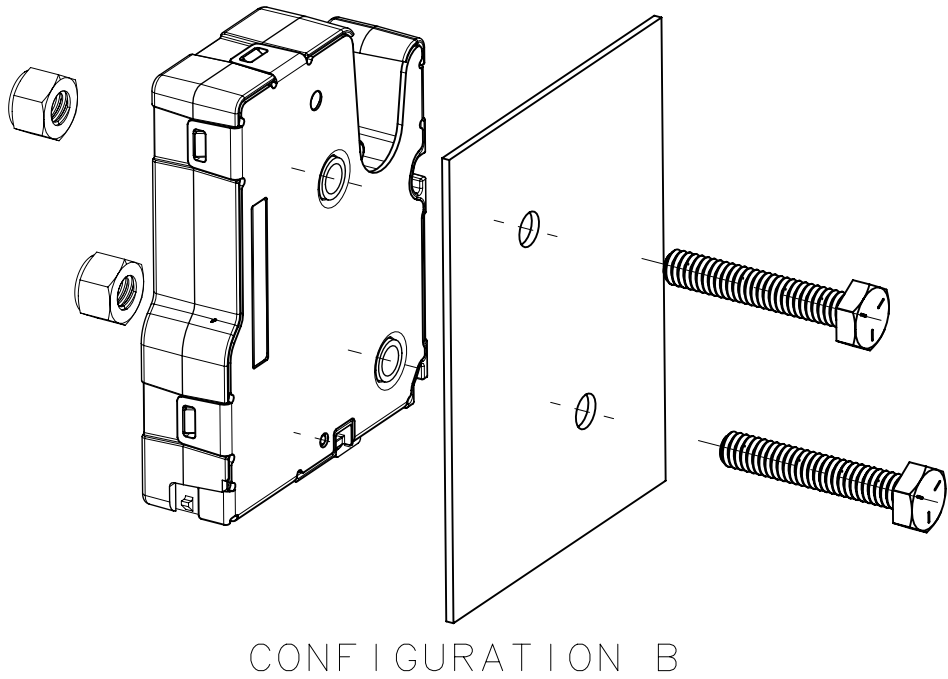
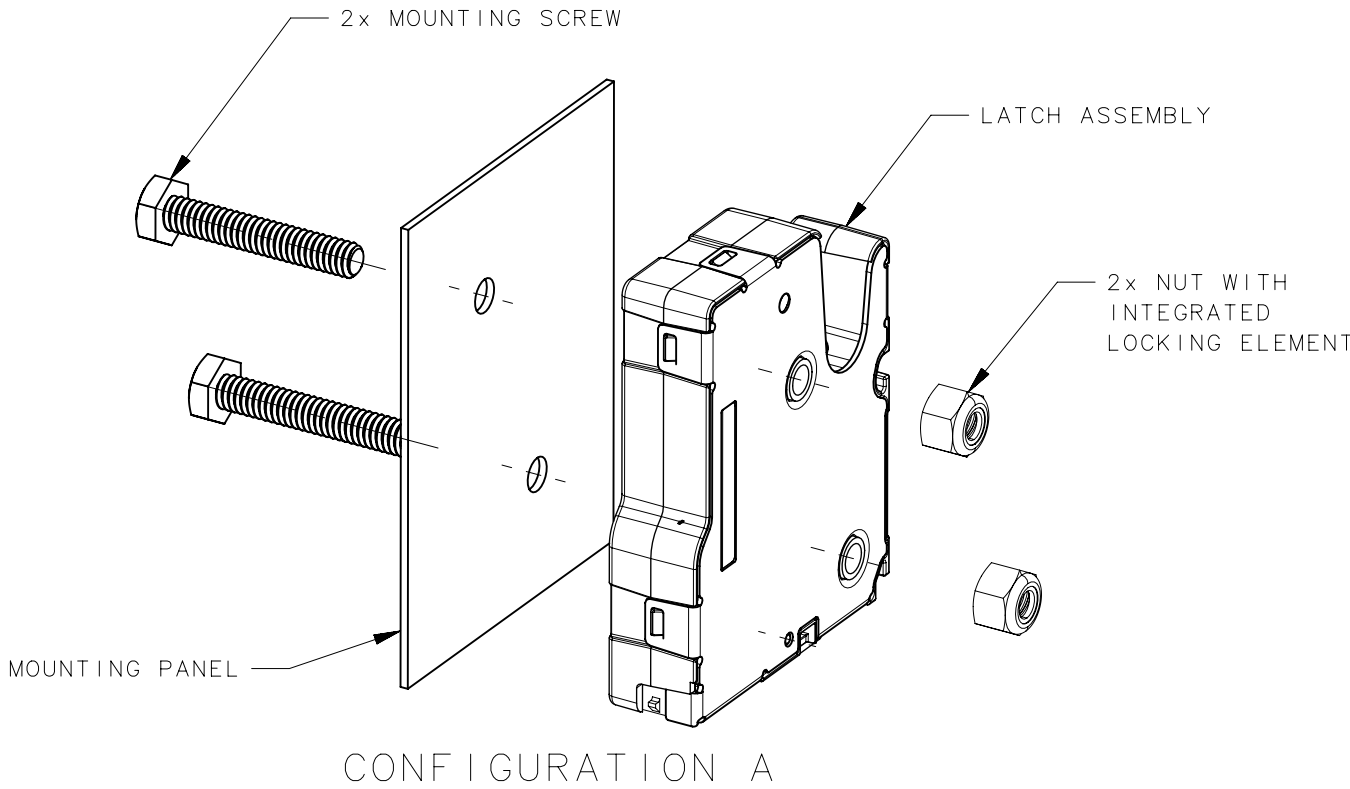
MANUAL OVERRIDE TRAVEL
DETAIL A
SCALE 2:1

DETAIL B
SCALE 2:1

SURFACE AREA N/A mm ² VOLUME N/A mm ³ PROPRIETARY ITEM EXCEPT FOR USES EXPRESSLY GRANTED IN WRITING, INFORMATION DISCLOSED HEREON IS CONFIDENTIAL AND ALL RIGHTS, PATENT AND OTHERWISE, ARE RESERVED BY SOUTHCO, INC.	THIRD ANGLE PROJECTION		
	MILLIMETERS [IN]		
	TOLERANCES UNLESS OTHERWISE NOTED	DESCRIPTION ELECTRONIC ACCESS SOLUTION R4-EM 9 SERIES DIE CAST	
	ALL DIMENSIONS WITHOUT TOLERANCES ARE FOR REFERENCE ONLY.	SIZE A3	SYSTEM NX
PER ASME Y14.5M-2009		DWG NO. J-R4-EM-9-2	SCALE 1:1
DRAWN BY BVH		DATE 06APR2017	SHEET 2 OF 4

MOUNTING CONFIGURATION
R4-EM-T1A5-1SC-P

REVISION HISTORY			
REV	DATE	BY	DESCRIPTION
F	21MAY2024	CRM/DMS	PRN: P2024-0871



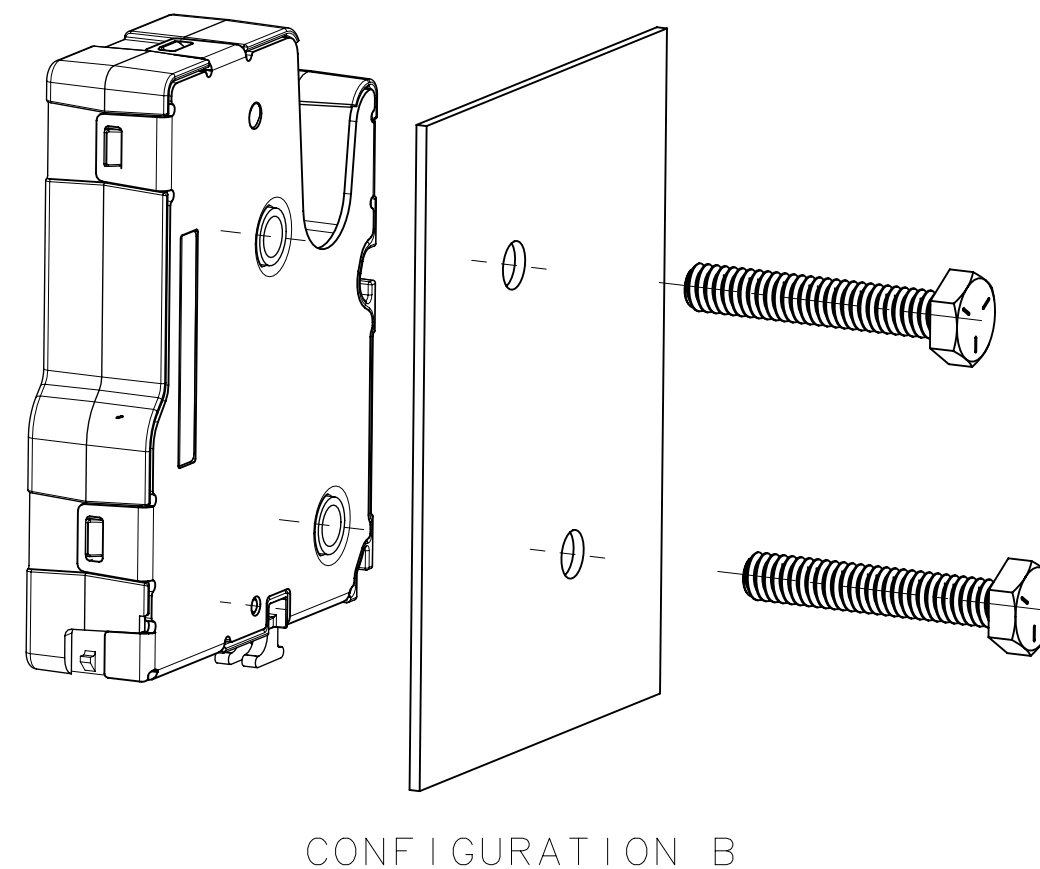
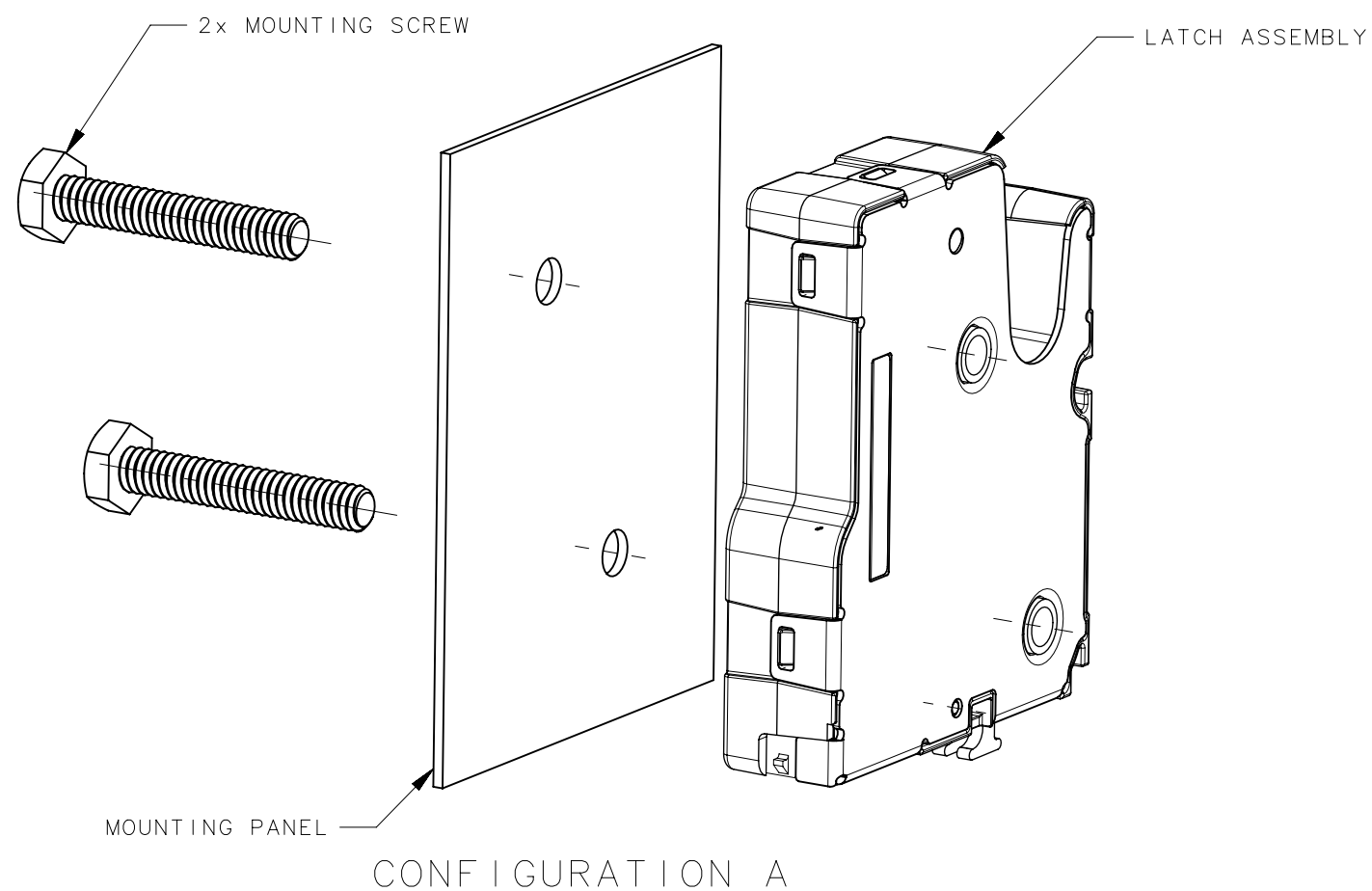
- NOTES:
- A. MAXIMUM RECOMMENDED FASTENER TORQUE, ALL CONFIGURATIONS: 5.0Nm.
- B. CONFIGURATION A AND D: USE OF FLAT WASHERS IS NOT RECOMMENDED. IF REQUIRED BY APPLICATION, DO NOT EXCEED MOUNTING TORQUE OF 4.0Nm.

SURFACE AREA N/Amm ² VOLUME N/Amm ³ PROPRIETARY ITEM EXCEPT FOR USES EXPRESSLY GRANTED IN WRITING, INFORMATION DISCLOSED HEREON IS CONFIDENTIAL AND ALL RIGHTS, PATENT AND OTHERWISE, ARE RESERVED BY SOUTHCO, INC.	THIRD ANGLE PROJECTION	southco® CONNECT • CREATE • INNOVATE		
	MILLIMETERS [IN]			
	TOLERANCES UNLESS OTHERWISE NOTED	DESCRIPTION ELECTRONIC ACCESS SOLUTION R4-EM 9-SERIES DIE CAST		
	ALL DIMENSIONS WITHOUT TOLERANCES ARE FOR REFERENCE ONLY.	SIZE A3	SYSTEM NX	DWG NO. J-R4-EM-9-2
PER ASME Y14.5M-2009		DRAWN BY BVH	DATE 06APR2017	SCALE 3:4
		SHEET 3 OF 4		

MOUNTING CONFIGURATION

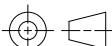
R4-EM-T2A5-1SC-P
AND
R4-EM-T3A5-1SC-P

REVISION HISTORY			
REV	DATE	BY	DESCRIPTION
F	21MAY2024	CRM/DMS	PRN: P2024-0871



NOTES:

A. MAXIMUM RECOMMENDED FASTENER TORQUE, ALL CONFIGURATIONS: 5.0Nm.

	THIRD ANGLE PROJECTION			 CONNECT • CREATE • INNOVATE			
	MILLIMETERS [IN]						
SURFACE AREA	N/Amm ²	TOLERANCES UNLESS OTHERWISE NOTED		DESCRIPTION			
VOLUME	N/Amm ³	ALL DIMENSIONS WITHOUT TOLERANCES ARE FOR REFERENCE ONLY.		ELECTRONIC ACCESS SOLUTION			
PROPRIETARY ITEM				R4-EM 9-SERIES DIE CAST			
EXCEPT FOR USES EXPRESSLY GRANTED IN WRITING, INFORMATION DISCLOSED HEREON IS CONFIDENTIAL AND ALL RIGHTS, PATENT AND OTHERWISE, ARE RESERVED BY SOUTHCO.				SIZE A3	SYSTEM NX	DWG NO. J-R4-EM-9-2	
	PER ASME Y14.5M-2009	DRAWN BY BVH		DATE 07SEP2018	SCALE 1:1	SHEET 4 OF 4	